

CLOSED SUBLATTICES OF $\ell_1(\Gamma)$ IN THE NORMING-M-BASIS HEREDITY PROBLEM

FA BANACH RUN PACKET

STATUS

Partial result. Closed sublattices of $\ell_1(\Gamma)$ have 1-norming Markushevich bases. Consequently, closed subspaces of $\ell_1(\Gamma)$ that are finite codimensional perturbations of closed sublattices have norming M-bases.

SOURCE PROBLEM

Hajek and Russo ask the following heredity question for norming Markushevich bases [2, Problem 6.1]. The problem appears in Section 6 of arXiv:2402.18442.

Problem 1 (Hajek–Russo). Let X be a Banach space with norming M-basis and let Z be a subspace of X . Must Z admit a norming M-basis as well?

The source identifies subspaces of $\ell_1(\Gamma)$, Γ uncountable, as the simplest possible counterexample territory and notes that it is not even known whether all such subspaces are Plichko. It also records that hyperplanes and closed sublattices of $\ell_1(\Gamma)$ are known to be 1-Plichko [2, Section 6].

IDEA OF THE PROOF

A closed sublattice L of $\ell_1(\Gamma)$ is much more rigid than a general closed subspace. Its norm is additive on positive vectors, so L is an AL-space. Kakutani’s representation theorem writes L as an $L_1(\mu)$ space. The coordinate functionals of the ambient $\ell_1(\Gamma)$, restricted to L , are lattice homomorphisms and separate points of L . In an $L_1(\mu)$ space, non-zero lattice homomorphisms live on atoms. Since the ambient coordinates separate points, there can be no nonatomic part. Thus L is an atomic L_1 space, hence a weighted $\ell_1(A)$. Its atom basis is the required 1-norming M-basis.

THE RESULT

Lemma 1. *Let L be a closed sublattice of $\ell_1(\Gamma)$. Then L is lattice isometric to $\ell_1(A, w)$ for some set A and weights $w_a > 0$. In particular, L is linearly isometric to $\ell_1(A)$.*

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Proof. The lattice operations on L are inherited coordinatewise from $\ell_1(\Gamma)$. If $u, v \in L_+$, then

$$\|u + v\|_1 = \sum_{\gamma \in \Gamma} (u_\gamma + v_\gamma) = \|u\|_1 + \|v\|_1.$$

Thus L is an AL-space. By Kakutani's representation theorem for AL-spaces, L is lattice isometric to $L_1(\mu)$ for some measure space (Ω, Σ, μ) ; see, for instance, the standard formulation quoted in [1, Theorem 4.27].

For each $\gamma \in \Gamma$, let

$$h_\gamma = \delta_\gamma|_L.$$

This functional is either zero or a lattice homomorphism: for example,

$$h_\gamma(x \vee y) = \max\{h_\gamma(x), h_\gamma(y)\}$$

and similarly for $x \wedge y$, since the operations are coordinatewise. The family $\{h_\gamma : \gamma \in \Gamma\}$ separates points of L .

Transport these functionals through the lattice isometry $L \simeq L_1(\mu)$. We use the elementary description of lattice homomorphisms on L_1 : a non-zero continuous lattice homomorphism on $L_1(\mu)$ is supported on a single atom of μ . Indeed, such a homomorphism is a positive functional, hence is represented by a positive measure absolutely continuous with respect to μ . If it gave positive mass to two disjoint measurable pieces, the identity $\varphi(f \wedge g) = \min\{\varphi(f), \varphi(g)\}$ applied to positive disjoint f, g would be violated. Therefore its representing measure is concentrated on one atom.

Since the transported coordinate homomorphisms separate points of $L_1(\mu)$, every non-zero $f \in L_1(\mu)$ must be detected by an atom. Hence the nonatomic part of μ is null. Therefore $L_1(\mu)$ is an atomic L_1 space, i.e. lattice isometric to a weighted $\ell_1(A, w)$. Rescaling the atom vectors gives a linear isometry with $\ell_1(A)$. $\square$

Theorem 1. *Every closed sublattice L of $\ell_1(\Gamma)$ admits a 1-norming Markushevich basis.*

Proof. By Lemma 1, L is linearly isometric to $\ell_1(A)$. The canonical unit vectors $(e_a)_{a \in A}$, together with the coordinate functionals $(e_a^*)_{a \in A}$, form a fundamental biorthogonal system. The linear span of the coordinates is 1-norming: for every $x \in \ell_1(A)$,

$$\|x\|_1 = \sup \left\{ \left| \sum_{a \in F} \varepsilon_a x_a \right| : F \subset A \text{ finite, } |\varepsilon_a| \leq 1 \right\}.$$

Thus the canonical system is a 1-norming M-basis, and transporting it through the isometry gives a 1-norming M-basis of L . $\square$

Corollary 1. *Let Z be a closed subspace of $\ell_1(\Gamma)$. Suppose there is a closed sublattice $L \subseteq \ell_1(\Gamma)$ such that, for $Y = Z + L$,*

$$\dim(Y/L) < \infty \quad \text{and} \quad \dim(Y/Z) < \infty.$$

Then Z admits a norming M-basis.

Proof. Theorem 1 gives a norming M-basis for L . The finite-codimensional invariance theorem promoted in the companion run packet [4] says that a Banach space has a norming M-basis if and only if each of its closed finite-codimensional subspaces does. Since Y/L is finite-dimensional, L is finite-codimensional in Y , so Y has a norming M-basis. Since Y/Z is finite-dimensional, Z is finite-codimensional in Y , and the same invariance theorem gives a norming M-basis for Z . $\square$

LIMITATIONS

The proof uses that L is a sublattice. Arbitrary closed subspaces of $\ell_1(\Gamma)$ do not inherit the coordinate lattice operations, and the argument gives no atomic representation for them. Thus the main subspace problem for $\ell_1(\Gamma)$, and hence Problem 6.1 in general, remains open.

NOVELTY AND LITERATURE CHECK

The source paper's arXiv page lists version 3, revised September 30, 2025, and still presents Problem 6.1 as open [3]. Web searches for the exact heredity problem, norming M-bases in subspaces, Kalenda's $\ell_1(\Gamma)$ Plichko question, and closed sublattices of $\ell_1(\Gamma)$ found no later full solution and no exact closed-sublattice norming-M-basis statement. The packet should be read as a standard Banach-lattice observation applied to the source problem, not as a claim that the arbitrary $\ell_1(\Gamma)$ subspace problem is settled.

REFERENCES

- [1] C. D. Aliprantis and O. Burkinshaw, *Positive Operators*, Springer, 2006.
- [2] P. Hajek and T. Russo, *Norming Markushevich bases: recent results and open problems*, arXiv:2402.18442.
- [3] P. Hajek and T. Russo, arXiv page for *Norming Markushevich bases: recent results and open problems*, <https://arxiv.org/abs/2402.18442>.
- [4] FA Banach run packet, *Finite-codimensional heredity for norming Markushevich bases*, runs/fa.banach_001/solutions/partial/2402.18442.finite_codim_heredity_norming_mbasis.